

THE MORTALITY GAP

The Paradox - Lower prevalence. Higher mortality. A divergence consistent with a profound diagnostic coverage gap.

28.5 per 100,000

Age-standardised diabetes mortality rate, Nigeria 2023
Nigeria's rate fluctuated between 28.5 and 31.7 over 23 years

78.7%

Of Nigerian diabetics are undiagnosed, invisible to the health system
8 in 10 outside the reach of care

3.2×

Higher mortality than Canada in 2023
Gap widened from 1.7× in 2000 as Canada's rate declined

Age-standardised prevalence

IDF Diabetes Atlas 11th Edition · 2024 · Adults 20–79

Nigeria
3.0%

Canada
7.7%

Higher recorded prevalence in Canada, higher mortality in Nigeria

Undiagnosed proportion

IDF Diabetes Atlas 11th Edition · 2024 · Crude proportion · Adults 20–79

Nigeria
78.7%

Canada
28.9%

Diagnostic coverage gap 2.7× wider than Canada's

Age-standardised mortality rate

IHME GBD 2023 · Our World in Data · Per 100,000 population

Nigeria
28.5%

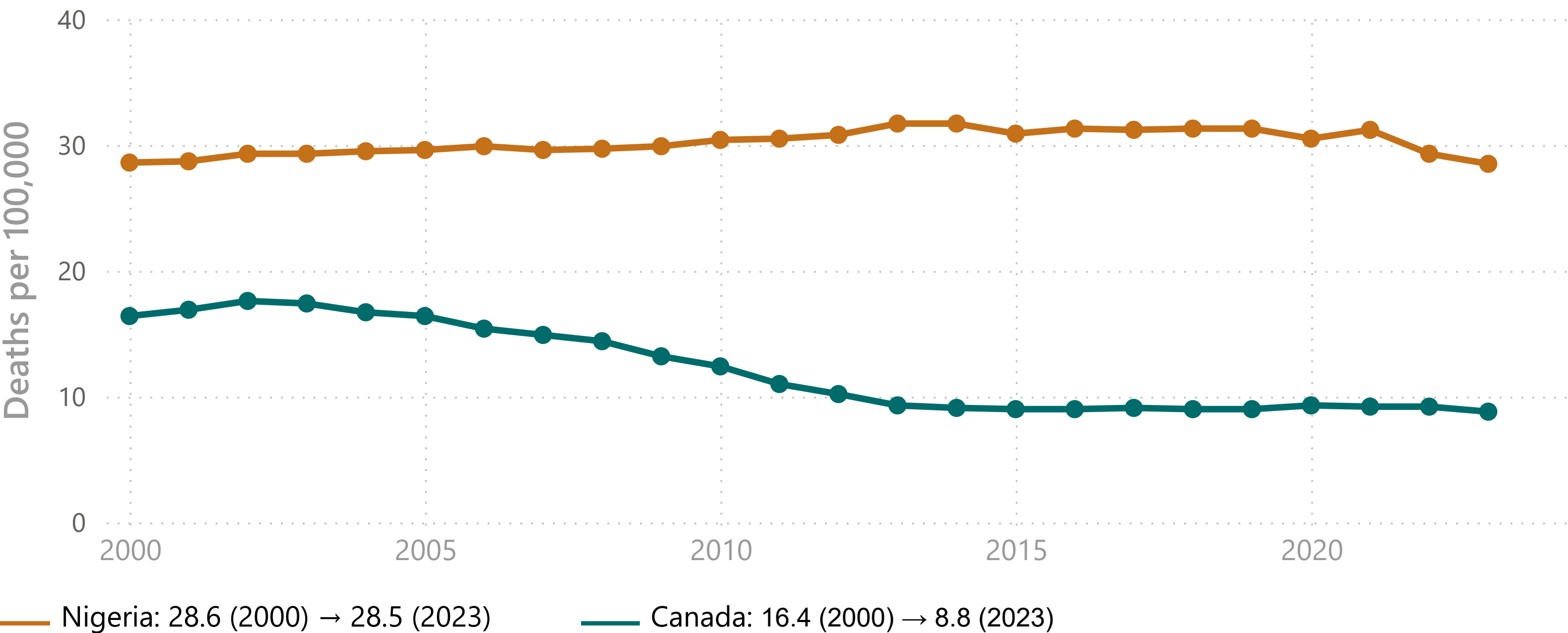
Canada
8.8%

Gap widened from 1.7× in 2000 to 3.2× in 2023

Age-standardised mortality rate trend

IHME GBD 2023 · Deaths per 100,000 population, age-standardised

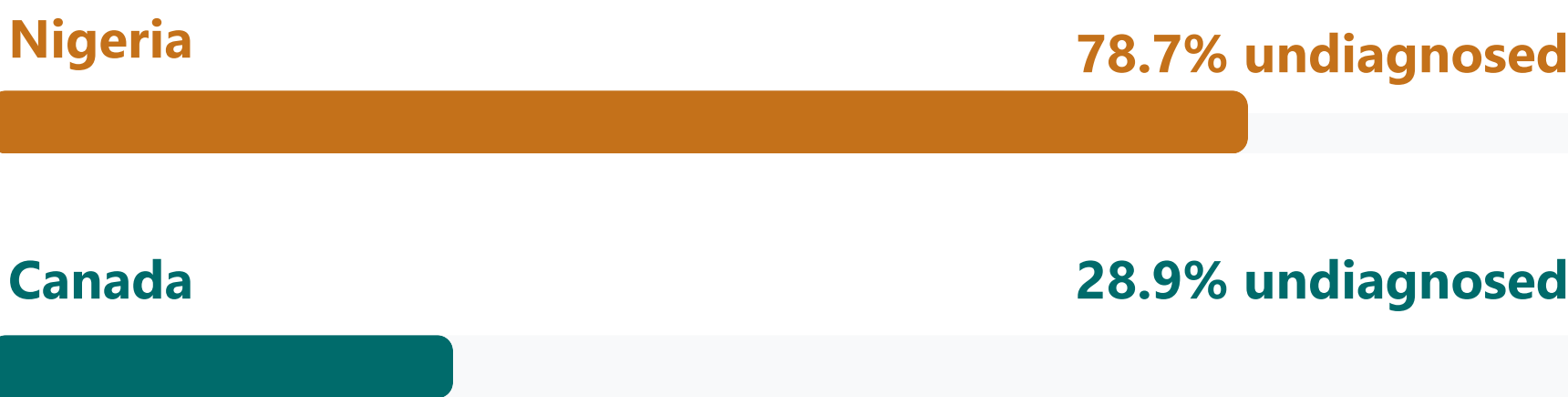
"Canada's age-standardised rate declined from 16.4 to 8.8 between 2000 and 2023. Nigeria's rate fluctuated over the same period, ending at 28.5."



Diagnostic coverage gap

IDF Diabetes Atlas 11th Edition · 2024 · Crude proportion · Adults 20–79

"Diagnostic coverage, not disease burden, distinguishes the two systems"



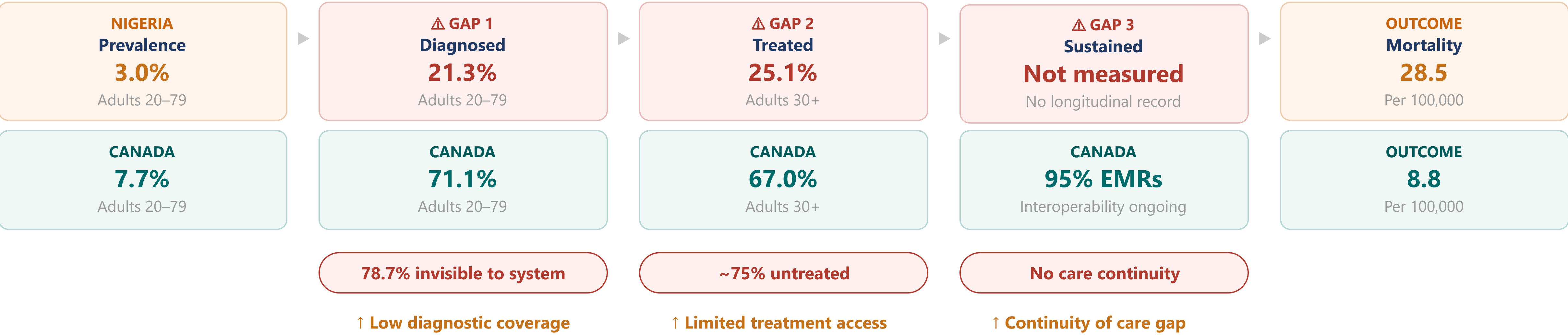
KEY INSIGHT:

Nigeria has significantly lower recorded diabetes prevalence than Canada, yet its age-standardised mortality rate is **3.2 times higher**, a ratio that has grown over 23 years as Canada's rate declined. This divergence points to a profound diagnostic coverage gap: **the majority of Nigeria's diabetes population remains invisible to the health system.**

Nigeria's diabetes care cascade where patients are lost

IDF 2024 · WHO GH0 2022 · IHME 2023 · CMA / Canada Health Infoway 2024

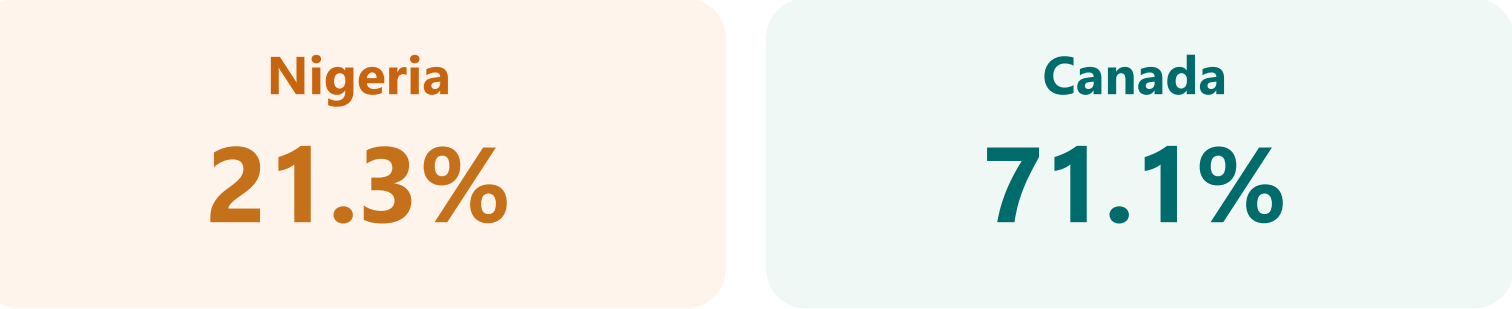
Three system-level gaps are associated with Nigeria's divergence from Canada across the care continuum



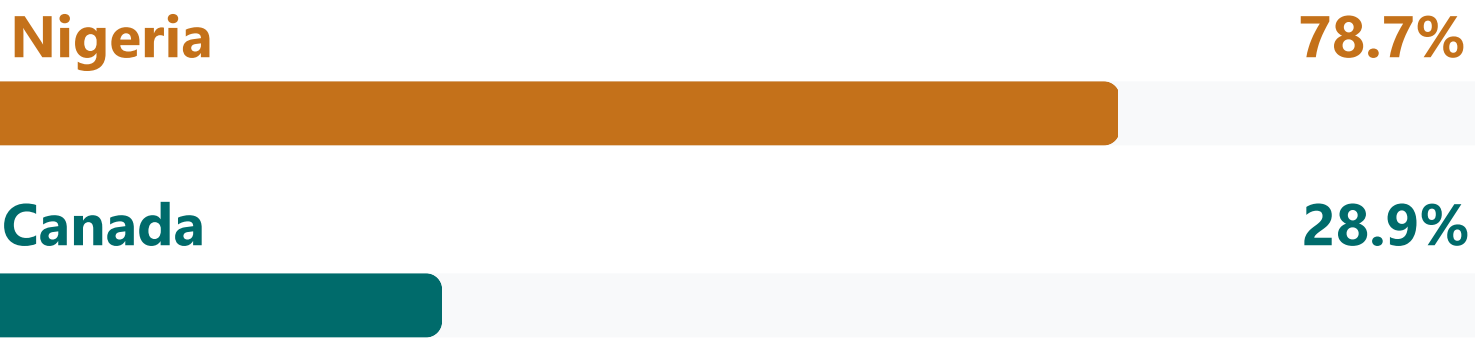
† Treated row uses Adults 30+ (WHO GH0 2022). All other rows use Adults 20–79 (IDF 2024). Denominators and reference years differ - cascade is indicative, not mathematically continuous. See Methods page.

Low diagnostic coverage
Invisible to the health system

"Fewer than 1 in 4 Nigerian diabetics are diagnosed"



UNDIAGNOSED PROPORTION



Diagnostic coverage gap 2.7× wider than Canada's

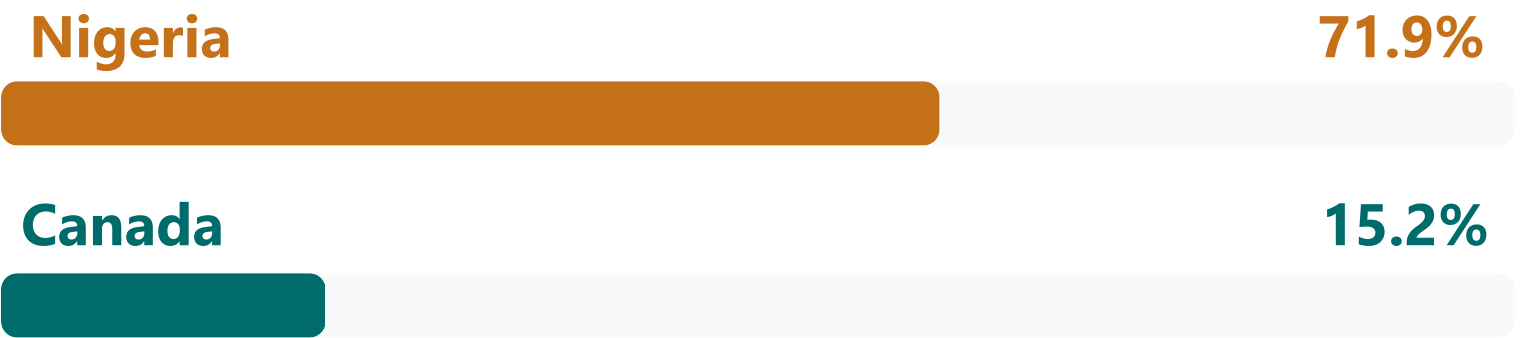
IDF Diabetes Atlas 11th Edition · 2024

Limited treatment access
Care is financially and structurally inaccessible at scale

"1 in 4 on treatment, 71.9% of health costs paid out of pocket"



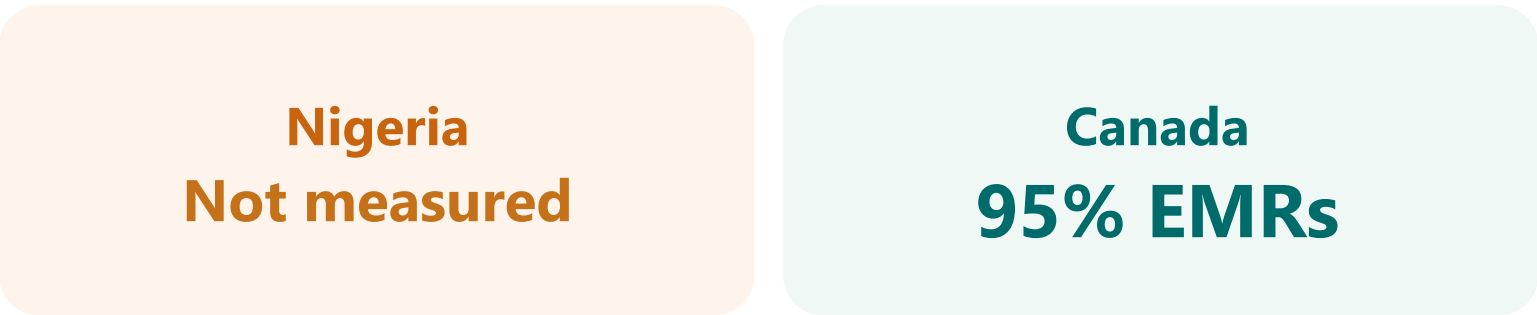
OUT-OF-POCKET EXPENDITURE



Two barriers prevent sustained treatment: financial inaccessibility (71.9% OOP, 28× lower health investment) and structural workforce constraints (3.8 vs 28.19 physicians per 10,000)

Continuity of care gap
No mechanism to retain or transfer patient information

"Canada tracks outcomes through 25 years of national surveillance; Nigeria has no equivalent longitudinal record"



NATIONAL DIABETES REGISTRY



No mechanism for Nigeria to track diabetes outcomes over time

CMA / Canada Health Infoway 2024 · PHAC · PMC6876649 · 2019

KEY INSIGHT: Nigeria's care pathway is associated with **three measurable system-level gaps**, in diagnostic coverage, treatment access, and continuity of care. These gaps are consistent with differences in health system performance, not differences in clinical knowledge or therapeutic options.

THE MORTALITY GAP

The Human Cost - When the system has no memory - Gap at the individual level

In Nigeria, a **substantial proportion of diabetes management occurs outside formal clinical settings**, led by family members without formal training, limited medication information, or systems that flag drug interactions. Without portable health records, every clinical encounter starts from zero. **The system has no memory of its patients.**

Diabetes deaths annually, Nigeria

40,990

Forty thousand families · IDF Diabetes Atlas 11th Edition · 2024

Nigerian diabetics with uncontrolled blood glucose, even when diagnosed

~2 in 3

The system finds them, but cannot sustain them · Iregbu et al. 2023

Of every healthcare cost in Nigeria paid directly out of pocket

71.9%

For many diagnosed diabetics, medication and monitoring are simply unaffordable · World Bank 2023

Clinical risk - Nigerian diabetes care

"Prescribing errors documented in Nigerian diabetes settings, consistent with LMIC patterns identified by WHO"

NIGERIA · PEER-REVIEWED

Prescribing errors, including medication strength omissions and incorrect dosage, are the most commonly documented error type in Nigerian diabetes outpatient care.

Gbadegesin & Adeyemi · Nigerian Journal of Pharmacy · 2024

GLOBAL · LMICS

In LMICs, approximately 80% of preventable medication errors occur at the prescribing stage, compared to 53% globally. Patients in LMICs bear a disproportionate burden of medication-related harm.

WHO Global Patient Safety Report · 2024

Infrastructure for safe, continuous care

SYSTEM ELEMENT	NIGERIA	CANADA
National diabetes registry	Not yet established	CCDSS since 1999
Medication reconciliation	No national standards	Accreditation Canada ROP - ISMP Canada Framework
Portable health record	No integrated system	95% using EMRs - Interoperability ongoing
CHEW diabetes protocol	No structured protocol	Regulated profession
Mortality trend	No surveillance data	2.1% annual decline

KEY INSIGHT: These system-level gaps are experienced directly at the individual level, **through discontinuity of care, incomplete clinical information, and avoidable risk**. Even among diagnosed Nigerian diabetics, approximately 2 in 3 have uncontrolled blood glucose. **The following page presents evidence-based recommendations that address detection, access, and continuity within Nigeria's context.**

Three evidence-based recommendations, each mapped to a gap identified on the preceding pages, each building on Nigeria's existing health infrastructure, and each aligned with WHO global frameworks.

DETECT
Enable population-level detection

Maps to: Low diagnostic coverage · National registry · NPHCDA infrastructure

CANADA · PHAC 2019

71.1%

Annual decline in diabetes mortality, associated with 20 years of systematic national surveillance

PMC6876649

NIGERIA'S ADAPTATION

A national diabetes registry building on Nigeria's existing National Diabetes Prevention and Control Programme and NPHCDA PHC expansion, enabling systematic detection at scale.

Nigeria's NPHCDA network and ongoing PHC expansion provide the infrastructure foundation.

Community Health Extension Workers (CHEWs) as frontline screening and registration agents.

WHO NCD Best Buys 2023 classifies this intervention as cost-effective in LMIC settings.

REACH
Extend treatment access to the unreached

Maps to: Limited treatment access · Community-based delivery · CHEW networks

NIGERIA VS CANADA · WHO GHO 2022

25.1% vs 67.0%

Population-level treatment coverage gap driven by two barriers: financial inaccessibility (71.9% OOP - World Bank 2023) and structural workforce constraints (3.8 vs 28.19 physicians per 10,000; WHO GHO SDG 3.c.1 · 2023)

WHO GHO · Adults 30+ · 2022

NIGERIA'S ADAPTATION

Structured community-based treatment delivery through CHEWs; addressing both barriers simultaneously: reducing geographic distance by bringing care into underserved communities, and reducing perpatient cost of access through a lower-cost community delivery model.

Nigeria's existing CHEW deployment programmes provide the workforce foundation for scaled delivery.

Nigeria's approximately 43,000 Community Health Extension Workers (CHEWs) as the primary delivery mechanism.

WHO NCD Best Buys 2023 classifies community-based NCD treatment delivery as cost-effective in LMIC settings.

CONNECT
Enable longitudinal patient tracking

Maps to: Continuity of care gap · Digital records · Mobile-first

CANADA · GARTNER / INFOWAY 2018

\$1B per year

Annual value generated by Canada's connected health information investment, including reduced duplicate testing, avoided adverse events, and improved chronic disease management

Connected Health Information in Canada - Benefits Evaluation Study · 2018

NIGERIA'S ADAPTATION

Digital health records at every PHC facility using mobile technology, enabling care continuity across providers at scale.

WHO NCD Best Buys 2023 classifies this intervention as cost-effective in LMIC settings.

KEY INSIGHT:
Detection, access, and continuity are interdependent. Nigeria's approximately 43,000 CHEWs and existing primary health care infrastructure represent a significant foundation upon which these recommendations can be built. **Realizing this potential requires governance frameworks, workforce training, and data protection safeguards.**

METHODOLOGY NOTES

1. Mixed reference years and trend interpretation

The ASMR trend uses real IHME GBD 2023 data. Nigeria: 28.6 (2000) to 28.5 (2023), fluctuating between 28.5 and 31.7. Canada: 16.4 to 8.8, a 46% decline. The mortality ratio widened from 1.74× to 3.24×.

2. Treatment coverage denominator

Treatment coverage is from WHO GHO for adults 30+ (2022). All other metrics are from IDF Atlas for adults 20–79 (2024). Each cascade stage is labelled with its specific source and population.

3. Undiagnosed proportion is crude

The undiagnosed proportion from IDF is a crude estimate, not age-standardised. The diagnosed proportion is derived as the complement(100 minus undiagnosed).

4. IHME modelled estimates

Nigeria's age-standardised mortality rate is a modelled estimate produced by IHME's Global Burden of Disease methodology, combining vitalregistration, survey, and verbal autopsy data with statistical modelling. Modelled estimates carry greater uncertainty in settings with less complete vital registration infrastructure.

5. Care cascade construction

The cascade is constructed from multiple sources with different denominators and reference years. It is indicative of system performance patterns, not a mathematically continuous pathway.

METHODOLOGY NOTES

6. Causal inference

This study demonstrates associations between health system indicators and mortality outcomes. It does not establish direct causation. The 2.1% annual decline in Canada is an observational trend, not a controlled experiment.

7. Medication safety evidence

Evidence draws on one Nigerian university health centre study and WHO global LMIC estimates. Findings are indicative, not nationally representative.

8. Community Health Extension Workers

Nigeria's CHEW workforce is estimated at approximately 43,000, based on peer-reviewed literature (Ballard et al., 2021). This figure may undercount more recent deployment.

9. Cost estimates

No specific peer-reviewed cost-per-unit figures were identified for the proposed recommendations. Costs are not quantified in this study. WHO NCD Best Buys 2023 cost-effectiveness classification is used instead.

10. Colour accessibility

Amber (#C4711A) on white achieves a contrast ratio of approximately 4.6:1, meeting WCAG 2.1 AA. All data elements are labelled with country names directly.

CONTRIBUTIONS

*This study synthesises epidemiological and health systems evidence to demonstrate how **diagnostic coverage, treatment access, and continuity of care** are associated with diabetes mortality outcomes in Nigeria relative to Canada, applying a comparative health systems framework to a question with direct relevance for health information infrastructure and earlier disease detection.*

COMPLETE SOURCE LIST

SOURCE	CITATION	PAGES
IDF Atlas 11th Ed	International Diabetes Federation. IDF Diabetes Atlas, 11th Edition. Brussels: IDF; 2025. Adults 20–79, 2024 estimates.	1, 2, 3, 4
IHME GBD 2023	Global Burden of Disease Collaborative Network. GBD Study 2023. Via Our World in Data.	1, 2, 3
WHO GHO - Treatment	World Health Organization. Global Health Observatory. Diabetes treatment coverage, age-standardised. 2022.	3, 5
WHO GHO - Physicians	World Health Organization. Global Health Observatory. SDG indicator 3.c.1. 2023.	3, 5
World Bank	World Bank. Out-of-pocket expenditure and health spend per capita, PPP. 2023.	3,4,5
PHAC - CCDSS	Public Health Agency of Canada. Twenty years of diabetes surveillance using the CCDSS. 2019. PMC6876649.	4, 5
Iregbu et al.	Iregbu S et al. Nigerian health care providers and diabetes self-management support. Sage Journals. 2023.	4
Gbadegesin & Adeyemi	Prescribing pattern and errors in diabetes mellitus at a Nigerian University Health Centre. Nigerian Journal of Pharmacy. 2024;58(2):213–218.	4
WHO Patient Safety	World Health Organization. Global Patient Safety Report 2024.	4
Accreditation Canada / ISMP	Accreditation Canada. Required Organizational Practices Handbook. 2021. ISMP Canada medication reconciliation framework.	4
CMA / Canada Health Infoway	2024 National Survey of Canadian Physicians. Leger. April–May 2024. n=1,145.	3, 4
Gartner / Infoway	Connected Health Information in Canada: A Benefits Evaluation Study. Gartner / Canada Health Infoway. 2018.	5
Ballard et al.	Compensation models for community health workers: comparison of legal frameworks across five countries. Journal of Global Health. 2021;11:04010.	5
WHO NCD Best Buys	WHO Global NCD Action Plan — best buys and recommended interventions. Updated 2022, endorsed 2023.	5

DISCLAIMER

This study represents an independent health informatics research study produced as a contribution to the evidence base on diabetes health system performance. It is not intended for clinical use or policy decision-making without further peer review. All data are sourced from publicly available authoritative sources cited throughout. Analysis reflects associations between health system indicators and mortality outcomes and does not establish causal mechanisms.